

OPEN RESEARCH DATA INFRASTRUCTURE FOR REPRODUCIBLE ANTIMALARIAL DRUG DISCOVERY: AI-ASSISTED WORKFLOWS AND COMPUTATIONAL PROVENANCE

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Reproducible computational drug discovery requires rigorous research data management (RDM), computational provenance, and open infrastructure. We present a data science framework supporting an integrated antimalarial pipeline comprising 65,856 compounds, 19,836 curated activity labels, GPU-accelerated molecular dynamics (16 systems $\times$ 10 ns), and multi-model machine learning benchmarks across five projects [1].

Our RDM infrastructure implements fail-closed validation gates, versioned artifacts with SHA256 hashes, and machine-readable audit trails recording parameters, software versions (GROMACS 2025.4, RDKit, PyTorch), and execution provenance for 15,000+ computational jobs. Each result links to frozen scripts and locked dependencies, enabling independent reproduction. Exploratory analyses are explicitly separated from validated results in hierarchical JSON manifests.

AI-assisted workflows accelerate prioritization through ensemble docking, graph neural networks (PyTorch Geometric GIN), and transformer encoders (ChemBERTa), while maintaining human-verifiable decision boundaries [2]. Python automation orchestrates SLURM HPC arrays, trajectory QC pipelines, and cross-project integration via version-controlled conda environments [3]. Quantum-inspired topological descriptors demonstrate honest-negative benchmarking where classical baselines outperform novel methods [4].

Open science practices include preregistered protocols, public deposition (Zenodo with reserved DOIs), FAIR metadata schemas, and transparent computational cost reporting (99.3% reduction through optimized grids). All infrastructure leverages open-source tools (OpenFF, scikit-learn, RDKit) deployable on modest HPC resources typical of African institutions. This framework provides a reusable template for computationally intensive workflows requiring reproducibility, provenance tracking, and responsible AI integration in resource-constrained environments.

- [1] M. V. Sao Temgoua et al., Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes, *J. Chem. Inf. Model.* (2026, in preparation).
- [2] M. V. Sao Temgoua et al., Graph Neural Networks for Scaffold-Aware Antimalarial Activity Prediction, (2026, in preparation).
- [3] M. V. Sao Temgoua et al., Quantum-Inspired Topological Representations for Molecular Discovery, (2026, in preparation).
- [4] M. V. Sao Temgoua et al., Pareto-Guided Monte Carlo Tree Search for Multi-Objective Drug Design, (2026, in preparation).